

AASTHA MALIK

Backend & Full-Stack Engineer | B.Tech CSE '28 | Delhi, India
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SUMMARY

Backend-focused software engineering student and product builder who has designed a Gmail-native AI agent for invoice recovery, shipped a multilingual video-captioning pipeline, and delivered two production client websites. Experienced with Python, FastAPI, PostgreSQL, React, Next.js, TypeScript, LLM integration, real-time updates, authentication, and media-processing workflows.

EDUCATION

B.Tech in Computer Science Engineering | 2024 - 2028 (Expected)
Guru Gobind Singh Indraprastha University (GGSIPU), Delhi

EXPERIENCE

Freelance Full-Stack Developer | Jun 2026 - Jul 2026 | Delhi, India | [SVJ Trust](#) | [Vedi Collections](#)

- Delivered two production client websites end to end, owning requirements, information architecture, responsive implementation, deployment, and post-launch technical support.
- Vedi Collections: Built and shipped an e-commerce platform with a Next.js frontend, FastAPI backend, Supabase PostgreSQL database, product catalog, order flow, admin panel, and Google Ads conversion tracking.
- SVJ Trust: Designed and launched a responsive public website spanning the founder's biography, blogs, video content, donations, an upcoming book, and contact flows.

SELECTED PROJECTS

Topline – Gmail-Native AI Revenue-Recovery Agent | Python, FastAPI, PostgreSQL, Gmail API, Gemini [GitHub](#)

- Designed a Gmail-native AI agent that triages inbox history in staged passes (list → header-only peek → transparent relevance scoring → full fetch only above threshold), extracting invoice facts from PDF and scanned attachments — with an OCR fallback — into a source-traceable ledger.
- Factored invoice state into two orthogonal Postgres columns (payment evidence vs. reminder eligibility) and enforced "no email can mark an invoice paid" at three independent layers — service logic, event ingestion, and a DB CHECK constraint — so only a verified Razorpay webhook or explicit owner override can confirm payment.
- Scoped Gemini to narrow, human-approved tasks only (drafting reminder emails, parsing plain-English approval replies) — never autonomous sending or payment decisions; built the Gmail/OAuth2 integration (Authlib, google-api-python-client) and a React 19 + TypeScript review dashboard.

Tendr – Gamified Productivity App | FastAPI, PostgreSQL, React 19, TypeScript

[GitHub](#) | [Live](#)

- Built 30+ REST endpoints for tasks, focus sessions, analytics, and a virtual-pet economy, enforcing priority-based XP rewards and daily overdue decay at the API layer.
- Implemented email/password authentication, six-digit OTP verification, Google OAuth2, and short-lived pending JWTs for new-user onboarding; independently shipped the application.
- Ranked #66 on Product Hunt on launch day.

Video Object Remover – AI Video Inpainting | SAM2, ProPainter, FastAPI

[GitHub](#) | [Demo](#)

- Connected click-to-select SAM2 segmentation with ProPainter video inpainting and streamed progress for 30-60 second inference runs using Server-Sent Events (SSE).
- Resolved 720p GPU memory failures with fp16 inference and chunked processing; deployed the application on Hugging Face Spaces and earned 6 GitHub stars.

CaptionFX – Multilingual Video Captioning Engine | Python, faster-whisper, ffmpeg, Gradio, React

captionfx.com | x.com/captionfx

- Building a multilingual video-captioning pipeline with per-language ASR routing — faster-whisper for English/European, a dedicated Hindi/Indic specialist model — instead of one multilingual model for everything, targeting stronger accuracy on Indian-language content than most tools offer.
- Chained Silero VAD → language ID → ASR → forced word-level alignment → caption shaping rules (~42 chars/line, ~15 chars/sec) → ffmpeg burn-in export via a temporary .ass file — the deliverable is a finished video, not a subtitle file.
- Deployed the ML pipeline as a GPU-backed Gradio app on Hugging Face Spaces and vetted every model's license for commercial use ahead of the paid product; frontend marketing site and editor UI in progress.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript

Backend: FastAPI, REST APIs, SQLAlchemy, Pydantic, Server-Sent Events (SSE), ffmpeg, Gmail API, LLM Integration (Gemini)

Frontend: React, Next.js, HTML, CSS

Data & Security: PostgreSQL, Supabase, OAuth2, JWT

Tools & Cloud: Git, Linux, Swagger/OpenAPI, Vercel, Render, Hugging Face Spaces, Razorpay